

DEBRE TABOR UNIVERSITY

Gafat Institute of Technology

Department of Computer Science

Internship Report

TITTLE: Software, Hardware, and Network Support at the Sekela Woreda Finance Office and ICT Directorate

Host Organization:Sekela Woreda Finance Office and ICT Directorate

Duration of Internship: Two Months

Prepared by: Endalew Matebie (ID: 6509)

Debre Tabor,

Ethiopia

Declaration and Approval

I am Endalew Matebie, Fourth-year Computer Science student Debre Tabor University, Gafat Institute of Technology, confirm that this internship report is based on my own work and experience during my practical training at the Sekela Woreda Finance Office and ICT Directorate.

Every system setup, hardware repair, technical configuration, and documentation included here comes from tasks I actually performed myself. Where I referred to external sources or existing materials, I have properly acknowledged them following standard academic practices.

Student Name: Endalew Matebie

Signature: _____

Date: _____

Approval Sheet

This internship report titled *"Computer Science Internship Report at* **Sekela Woreda Finance Office and ICT Directorate***"* has been submitted to the Department of Computer Science, Faculty of Computing and Informatics, **Debre Tabor University**, for evaluation and approval.

Company Supervisor: Mr. Adane

Signature: _____

Date: July 01-August 30 E.C

Academic Advisor: Mrs.Emebet

Signature: _____

Date:___August20-25 E.C

Acknowledgment

"I would like to express my sincere appreciation to the management and technical team at the Sekela Woreda Finance Office and the ICT Directorate for allowing me to gain practical experience with their network systems, databases, and maintenance equipment.

I am especially grateful to my on-site supervisors and colleagues. Their patient guidance, respect, and supportive environment made a significant difference in my learning. I am also deeply honored by their positive feedback and the potential job opportunity considered for me after graduation.Finally, I extend my gratitude to the Department of Computer Science at Debre Tabor University (Gafat Institute of Technology). Thank you for organizing this internship program and helping us bridge the gap between academic theory and real-world professional practice."

Executive summary

"This report covers my two-month practical internship at the Sekela Woreda Finance Office. The internship provided a great opportunity to bridge the gap between academic Computer Science theories and real-world IT operations.

During my time there, I gained hands-on experience tackling actual operational and technical challenges. On the software side, I fixed critical flaws in the legacy Payroll and IBEX systems. Furthermore, I built a new integrated search module that allows staff to quickly look up employee recordsI also developed a search function that helped staff easily find employee records using PIN or TIN numbers. In addition, I helped maintain office computers by upgrading workstations and replacing damaged RAM and power supply units (PSUs). I also worked on improving the office network infrastructure to provide better connectivity and support daily office activities.

I upgraded outdated workstations running unpatched Windows versions and replaced faulty RAM and PSUs. I also optimized the physical network infrastructure by installing network cabinets, terminating Cat6 UTP cables onto patch panels (using the T568B standard), and configuring Cisco managed switches with separate VLANs for different departments to enhance security and reduce broadcast domains."

List of Acronyms and Abbreviations

VoIP …… Voice over Internet Protocol

API …….. Application Programming Interface

Cat5e ……..Category 5e Cable

Cat6 ……. Category 6 Cable

Cat6a ……Category 6 Cable Augmented

CLI ……. Command-Line Interface

E.C. ……Ethiopian Calendar

EMI …… Electromagnetic Interference

FDF ……Fiber Distribution Frame

IBEX …….Integrated Budget and Expenditure System

ICT ……..Information and Communication Technology

IDC ……..Insulation Displacement Contact

IP …… Internet Protocol

IT …….. Information Technology

LAN …….Local Area Network

LCD ……. Liquid Crystal Display

LED ……. Light-Emitting Diode

MAC ……. Media Access Control

OS…….. Operating System

OWASP …….Open Worldwide Application Security Project

PC …….Personal Computer

PDU ……..Power Distribution Unit

PHP …… Hypertext Preprocessor

PIN …….Personal Identification Number

PoE ……Power over Ethernet

PSU ……..Power Supply Unit

RAM ……Random Access Memory

RegEx …….. Regular Expression

RFI ……. Radio Frequency Interference

RJ45 ……..Registered Jack 45

SQL …….. Structured Query Language

T568B ……Telecommunications Industry Association Standard 568B

TIA/EIA …..Telecommunications Industry Association / Electronic Industries Alliance

TIN …… Tax Identification Number

UPS ……Uninterruptible Power Supply

USB ……. Universal Serial Bus

VLAN ……Virtual Local Area Network

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CHAPTER ONE

PROJECT SELECTED AND WORKED OUT

1.1 Project Title and Summary

1.1.1 Project Report on Hardware Maintenance, IBEX Support, Network Cabling, and Payroll Application Development.

Project Summary:

During my internship as a Computer Science student at the Sekela Woreda Finance Office, I was assigned to the **ICT Directorate**. My project focused on upgrading both the software and hardware environments of the organization. The office depends on important financial systems, especially the **IBEX system**, to carry out its daily operations. Because of this, my main goal was to make sure these systems ran smoothly, stayed secure, and that the network remained stable and reliable.

The project was divided into two major parts:

1. **Software & Database Optimization**

I fixed serious data-entry problems in the office's old **Payroll system**. The system was allowing users to type in special characters that could damage the database. To solve this, I wrote **PHP Regular Expressions (RegEx)** that block these characters before they can cause any harm — keeping the database clean and safe.

2. Hardware & Network Infrastructure

I reorganized and rebuilt the office's messy physical network. This included:

- Installing a **central network cabinet** to organize all equipment in one place
- **Terminating Cat6 cables** onto patch panels for a clean, professional setup
- **Configuring Cisco managed switches** with **VLANs** to separate and secure network traffic
- **Updating vulnerable workstation operating systems** to close security gaps

1.2 Problem Statement and Justification

1.2.1 Problem Statement

Before this project began, the ICT environment at the Finance Office was facing two serious problems:

- **Software Level:**
  The old **Payroll** and **IBEX** systems had weak input validation. This meant users could accidentally type special characters (like @, #, or $) into fields that should only contain employee names. When this happened, it **corrupted records in the MySQL database**, and in some cases, the system **crashed during monthly financial audits** — causing delays and confusion.

- **Network & Hardware Level:**
  Office computers were running **outdated, unpatched versions of Windows**, making them easy targets for malware and security threats. On the physical side, the network cables were **disorganized and plugged directly into unmanaged switches** — with no central patch panels. This caused **frequent cable shorts and network disconnections**, even affecting regional branches.

1.2.2 Justification

Fixing these problems was essential. By adding **strict PHP RegEx validation**, we made sure only clean, correct data enters the database. This makes **salary calculations error-free** and saves the audit team hours of manual corrections.

At the same time, reorganizing the network with **patch panels, Cat6 cables, and VLAN configurations** ensures that sensitive financial data stays **secure**, unnecessary broadcast traffic is **reduced**, and daily operations run **smoothly — without network downtime**.

1.3 Objectives of the Project

1.3.1 General Objectives

To apply computer science principles to solve real-world organizational problems by enforcing software data integrity and building a highly reliable, secure IT and network infrastructure.

1.3.2 Specific Objectives

To achieve the general objective, the project was broken down into the following specific tasks:

- **Develop an integrated SQL search module** that allows staff to quickly find employee profiles using their **PIN** or **TIN**.
- **Perform preventive maintenance on office workstations** — including OS formatting and malware removal — to keep vulnerable computers secure and running properly.
- **Physically install network cabinets and route Cat6 UTP cables** to build a clean, organized, and reliable network infrastructure.
- To correctly terminate network cables onto patch panels using the standard T568B color coding.
- **To configure Cisco switches via Command-Line Interface (CLI) and assign Virtual LANs (VLANs)** to separate the Payroll department's traffic from the rest of the network.

1.4 Methodology

To complete this project successfully, I followed a **structured, step-by-step IT deployment methodology**. Each phase built on the previous one, ensuring the work was organized, tested, and reliable

1. Site Assessment & Requirement Gathering

I interviewed the finance staff to understand why the IBEX software kept crashing. I also carried out a physical survey of the server room and cable routes to map out the existing setup and identify problem areas.

2. System Design

I designed the PHP validation logic and sketched a logical network topology to plan where VLANs would be needed for better traffic control and security.

3. Hardware Installation

I wall-mounted the network rack, routed the Cat6 cables, and used a **punch-down tool** to terminate the wires into the patch panel — creating a clean, professional network setup.

4. Testing and Handover

I ran software unit tests to confirm the database correctly rejected invalid symbols. I also used **LCD cable testers** to verify end-to-end physical network connectivity. Once everything passed testing, I handed the upgraded system over to the staff.

1.5 Literature Review

1.5.1 Input Validation & Data Integrity

According to **OWASP (Open Worldwide Application Security Project)** guidelines, improper input validation is one of the leading causes of software vulnerabilities — including **SQL Injection** and database corruption. Enforcing strict rules at the application layer, such as using **Regular Expressions** to accept only alphabetical characters in name fields, ensures that the underlying database stays **clean, accurate, and trustworthy** for financial reporting.

1.5.2 Network Segmentation and VLANs

Kurose and Ross (2021), in Computer Networking: A Top-Down Approach, explain that implementing **Virtual Local Area Networks (VLANs)** allows network administrators to logically segment a network. By placing the Payroll department on its own VLAN, **broadcast traffic is significantly reduced**, bandwidth is optimized, and sensitive financial data is **secured from unauthorized access** by other departments.

1.5.3 Structured Cabling Standards

The **ISO/IEC 11801** and **TIA/EIA-568** standards require that enterprise networks use **structured cabling**. Following these standards means using **central patch panels** and the **T568B pinout standard** instead of connecting PCs directly to switches.

This approach:

- **Minimizes physical strain** on network equipment
- **Reduces Electromagnetic Interference (EMI),**and
- **Makes future troubleshooting significantly faster** and more organized

CHAPTER TWO

BACKGROUND OF THE HOSTING COMPANY

2.1 Brief History of the Organization

The **Sekela Woreda Finance Office** is a key government institution with the core mandate of managing the economic and financial resources of the Woreda — making sure public funds are properly **collected, budgeted, and distributed**.

Over the years, as the Woreda's population and administrative needs grew, the office expanded its **technological and human resource capacities** to meet the growing demands of local administration.

2.2 Technological Improvement and Digital Systems

- A major milestone in the office's modern history was the integration of **Information Communication Technology (ICT)** into its daily operations. This digital transformation introduced several key systems:
- **IBEX System** — the core financial management platform used for budgeting, accounting, and reporting.
- **Payroll System** — used for processing employee salaries and deductions accurately and on time.
- **Integrated Database (MySQL)** — stores employee records, financial data, and audit information securely.

These systems significantly improved the office's **efficiency, accuracy, and transparency** in managing public funds. However, as the office continued to grow, the need for **stronger input validation, upgraded hardware, and a structured network infrastructure** became clear — which led to this internship project.

2.3 Vision, Mission, and Core Values

Vision:

To be a model financial institution that ensures **transparent, accountable, and efficient management of public resources** for sustainable development within the region.

Mission:

To provide quality financial services through:

- Effective budget management
- Accurate revenue collection
- Timely payroll processing
- Rigorous financial auditing

…there by contributing to good governance and comprehensive community development.

Core Values:

Core Value	Description
Accountability	Taking absolute responsibility for all financial transactions and decisions.
Transparency	Maintaining open, clear, and auditable communication regarding public funds.
Integrity	Upholding honest and ethical conduct in all data handling and financial processing.
Efficiency	Ensuring the optimal use of resources, including time and IT infrastructure.
Teamwork	Fostering collaboration and mutual support across all departments.

Table 2.1 Core Values of Technological Improvement

2.4 Main Products and Services

As a government entity, the office does not manufacture physical products. Instead, it provides critical **financial and administrative services**:

1. Financial Management & Budgeting

Planning, allocating, and distributing municipal budgets across local government sectors.

2. Payroll Processing

Managing monthly salary calculations, pension deductions, and tax withholdings for all public servants in the Woreda using the digital **IBEX system**.

3. Taxation & Auditing Services

Maintaining precise taxpayer databases, verifying identification through **Personal Identification Numbers (PIN)** and **Tax Identification Numbers (TIN)**, and conducting rigorous annual financial audits.

4. ICT Infrastructure Support

Providing the underlying digital architecture — networks, databases, and servers — that ensures secure data storage and communication across all regional branches.

2.5 Main Customers and End Users

The primary end-users of the office's services include:

1. Government Employees

Who depend completely on the office's payroll system for accurate and timely monthly salary processing.

2. Taxpayers and Local Businesses

Who interact with the office for tax assessments, TIN registrations, and financial clearance certificates.

3. Regional Government Bodies

Who rely on the accurate digital financial reports generated by the auditing department to monitor local economic health.

2.6 Organizational Structure

The Sekela Woreda Finance Office operates under a hierarchical structure to maintain order and accountability among its 60 employees.

- Woreda Finance Head: The highest executive officer overseeing all branches.
- Finance & Budget Directorate: Handles IBEX data entry, local expenditures, and payroll management.
- Tax & Revenue Directorate: Manages audits and taxpayer databases.
- ICT Directorate: Responsible for system maintenance, network uptime, software troubleshooting, and database security. (This is the specific department where my internship took place).

➢

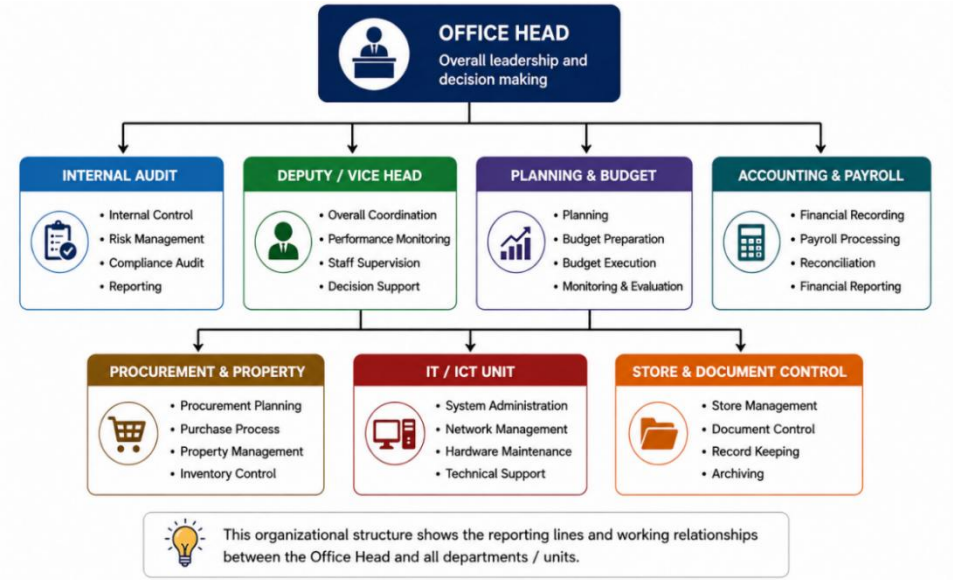


Figure 2.1 Organizational Structure

2.7 Work Flow of the Organization

Here is a clear, step-by-step breakdown of how the organization's workflow operates, translated into everyday language:

2.7.1. Getting the Money (The Budgeting Phase)

What happens: Everything starts when the regional government allocates and sends funds to the organization.

2.7.2 Managing Money & Collecting Taxes (Finance & Revenue)

Once the funds arrive, two departments go to work simultaneously:

- **The Finance Department:** Handles day-to-day expenses and monthly staff salaries using a specialized financial software called **IBEX**.
- **The Revenue Department:** Collects local taxes from citizens or businesses, issuing digital receipts tied to unique tax identification numbers (**TIN**).

2.7.3.Storing Everything Safely (ICT & Network)

- **What happens:** All the digital receipts, payments, and payroll records travel across the office's local computer network (**LAN**). They are safely stored in a centralized database

managed by the Information and Communication Technology (**ICT**) Directorate so they don't get lost.

4. Checking the Books (The Auditing Team)

What happens: At the end of the process, the **Auditing Team** reviews all the stored data. Using digital search tools (like SQL queries and PIN/TIN lookups), they verify that every transaction is legal, correct, and compliant, ultimately creating the final fiscal year reports.

CHAPTER THREE

OVERALL INTERNSHIP EXPERIENCE AND SPECIFIC WORK

3.1. Company Selection & Section Assignment

Reason for Company Selection

I chose Sekela Woreda Finance Office for our internship because it offered the perfect mix of real-world IT challenges. The office relies heavily on critical financial systems like IBEX, operates across multiple branches that need to stay connected, and deals with complex database audits—all of which gave us a chance to apply our computer science skills in meaningful ways.

Role of Computer Science in Finance Departments

Computer science professionals play an essential role in modern finance offices. I help digitize manual accounting processes, protect sensitive financial databases from breaches or tampering, keep networks running smoothly across branches, automate tax and payroll calculations, and troubleshoot enterprise systems like IBEX when issues arise. Without IT support, financial operations would be slower, less accurate, and far more vulnerable to errors and security threats.

3.2 Executed Work Tasks & Technical Details

Software Validation, Auditing & PIN Search Development

One of our first tasks was tackling a frustrating issue in the legacy Payroll and IBEX systems—they allowed users to type just about anything into text fields, including symbols and numbers where only names should go. This was causing all sorts of data problems behind the scenes.

To fix this, I wrote input validation rules using Regular Expressions (RegEx) in PHP. Essentially, we programmed the system to strictly accept only letters and appropriate characters in name fields, rejecting anything that didn't belong there.

I also built custom SQL search queries and reporting modules that made it much easier for staff to audit employee records. Now, instead of digging through piles of data manually, they can quickly look up any employee using their PIN or Tax Identification Number (TIN) and pull up complete profiles in seconds.

3.3 Workstation OS & Preventive Maintenance

I tackled serious security risks on office computers that was running outdated, unpatched Windows versions. Many of these machines were vulnerable to malware and other threats, so I took a comprehensive approach to fix them.

For each workstation, we performed fresh operating system installations to start from a clean slate, applied all the latest security patches, installed reliable antivirus software, and tweaked system boot configurations to speed up startup times. We also replaced failing power supply units (PSUs) and faulty RAM modules to prevent random crashes and hardware failures.

3.4 Network Cabling, Rack Setup & Switch Configuration

On the networking side, I started by mounting wall-rack enclosures and organizing patch panels using standard rack unit spacing to keep everything neat and accessible.

3.5 Network Cabling Components

Network cables are used to connect and transfer data and information between computers, routers, switches and storage area networks. These cables are essentially the carrier or media through which data flows.We can call it as the physical backbone of wired communication networks, providing
reliable, low-latency, and high-throughput transmission of data between devices.

1. Fiber Optic Patch Cord (Top-Left, Yellow Cable)

Name: Single-mode (or Multi-mode) Fiber Optic Patch Cable / Jumper.

Function: Transmits data using light signals over optical fibers rather than electrical pulses over copper.

Key Roles in a Report:

- **High-Speed & Long-Distance Data Transfer:** Provides significantly higher bandwidth and lower attenuation compared to traditional copper cables.
- **Interconnecting Network Hardware:** Connects optical transceivers (SFP/SFP+ modules) installed in network switches, routers, servers, or Fiber Distribution Frames (FDFs).
- **Immunity to EMI:** Completely immune to Electromagnetic Interference (EMI) and Radio Frequency Interference (RFI), making it ideal for environments with heavy electrical noise.

2. Ethernet / RJ45 Patch Cables (Middle, Bottom-Left, & Right Packages)

- **Name:** Twisted-Pair Copper Patch Cable (Cat5e / Cat6 / Cat6a with RJ45 connectors).
- **Function:** Transmits data via electrical signals through twisted pairs of copper wires.

Key Roles in a Report:

- **Local Area Network (LAN) Connectivity:** Connects end-user devices (computers, IP phones, access points) to wall outlets or patch panels.
- **Short-Distance Switch Connections:** Bridges connections between patch panels, Ethernet switches, routers, and firewalls within a server rack or telecommunications closet.
- **Power delivery (PoE):** Supports Power over Ethernet (PoE) to supply both power and data to devices such as IP cameras, VoIP phones, and wireless access points without extra power adapters.



Figure 3.1 Ethernet Cables

Table 3.1 Network Cabling Components

Component	Signal Type	Primary Use Case	Key Advantage
Fiber Optic	Light /	Uplinks, server-to-switch	High speed, high bandwidth,

Patch Cord	Optical	links, long runs	immune to EMI
Ethernet Patch Cable	Electrical	LAN connectivity, desktop-to-wall links, PoE	Cost-effective, standard interface (RJ45), simple deployment

3. Cabinet Box

A network **cabinet box** (also known as a server rack, network enclosure, or rack enclosure) is a standardized frame or cabinet designed to house, organize, and protect critical networking hardware and patch panels.

Primary Functions in a Network Setup

- **Hardware Organization & Space Efficiency:** Standardizes equipment mounting using "Rack Units" . It allows network switches, patch panels, routers, firewalls, and servers to be stacked vertically, minimizing physical footprint.
- **Cable Management:** Provides structured channels, routing rings, and cable trays to neatly route incoming network cables and patch cords. This prevents tangled lines, reduces cable strain, and makes tracing and replacing cables straightforward during troubleshooting.
- **Physical & Environmental Protection:** Encloses sensitive IT hardware to protect it from dust, physical damage, unauthorized access, and accidental unplugging. Lockable doors and side panels provide essential physical security for enterprise networks.
- **Thermal Management:** Facilitates proper airflow through integrated ventilation slots, mesh doors, and top-mounted cooling fans, preventing equipment overheating and ensuring stable network performance.
- **Centralized Power Distribution:** Holds Power Distribution Units (PDUs) and Uninterruptible Power Supplies (UPS) to streamline electrical cabling and maintain backup power for connected equipment.

4.Patch Panel

A patch panel is a central hardware assembly mounted in a network rack that contains multiple network ports to connect, organize, and manage incoming and outgoing Ethernet or

fiber optic cables.

"It can be reused where appropriate and when it remains in good working condition." highlights its primary organizational benefit and potential for reuse in different configurations.



Figure 3.3 Patch Panel

5. Switch

A network switch is a device that connects computers, printers, servers, and other network devices in a LAN. It enables communication between connected devices and can use VLANs to separate different departments or network groups.

Example: Cisco Catalyst 2960-24TT is a managed switch that supports VLAN configuration and network management.

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname Finance_Switch
Finance_Switch(config)# vlan 30
Finance_Switch(config-vlan)# name Payroll_Dept
Finance_Switch(config-vlan)# exit
Finance_Switch(config)# interface fastEthernet 0/1
Finance_Switch(config-if)# switchport mode access
Finance_Switch(config-if)# switchport access vlan 30
Finance_Switch(config-if)# exit
Finance_Switch(config)# interface vlan 30
```

Finance_Switch(config-if)# ip address 192.168.30.10 255.255.255.0

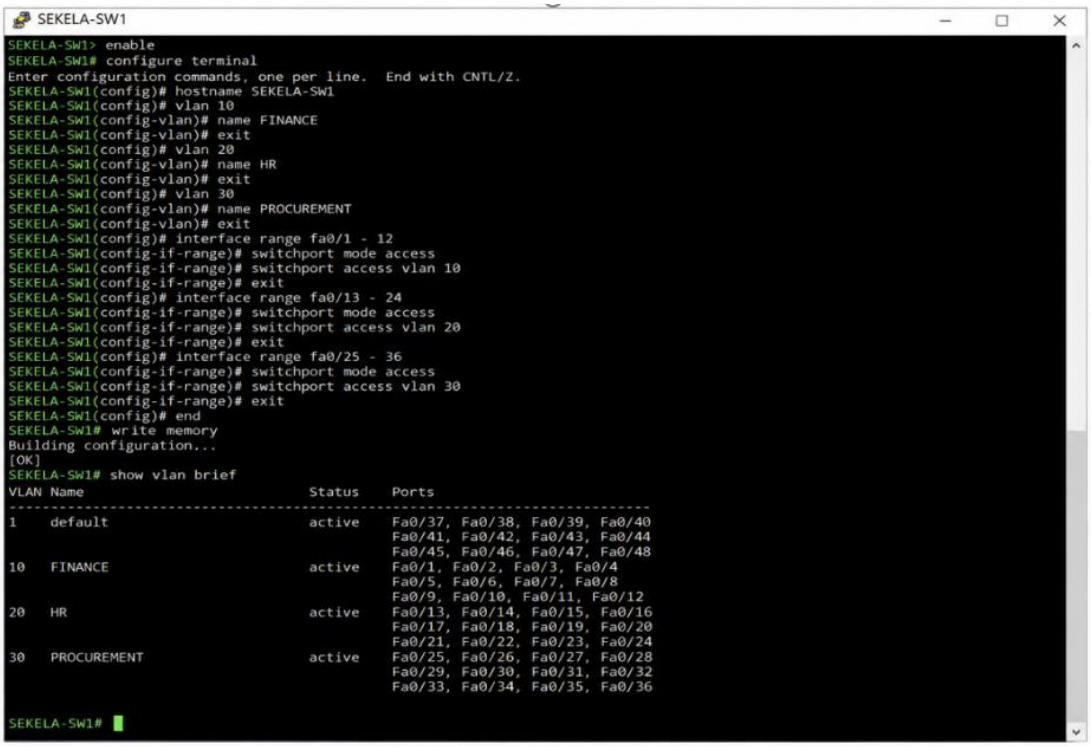
Finance_Switch(config-if)# no shutdown

Finance_Switch(config-if)# exit

Finance_Switch(config)# ip default-gateway 192.168.30.1

Finance_Switch(config)# do write

- This setup created a dedicated VLAN (Virtual Local Area Network) for the Payroll Department, isolating their network traffic for better security and performance.



3.6 Hardware and OS Preventive Maintenance

To resolve frequent system crashes across several office computers caused by hardware degradation and malware infections, I performed full physical and software overhauls:

- **Hardware Servicing & Component Replacement:** I opened the computer cases, used compressed air to safely clear heavy dust buildup from motherboards and cooling components, and replaced failing Power Supply Units (PSUs) and defective RAM modules to restore system stability.
- **Software Clean Installation & Security Enhancement:** On the software side, I wiped and formatted infected hard drives, reinstalled clean and updated Windows operating systems, and deployed centralized Endpoint Antivirus software to protect the computers against future malware threats.

Key Highlights of This Section:

- **Problem-Solution Structure:** It clearly outlines the initial issue (crashes due to hardware and malware) and details your step-by-step technical intervention.
- **Professional Terminology:** Uses clear, industry-standard language (*hardware servicing*, *system stability*, *clean installation*, *endpoint security*) that effectively highlights your technical competence.

3.7 Engineering Tools & Equipment Utilized

Table 3.2 Tool/Equipment

Tool / Equipment	Core Technical Function	Practical Internship Application
Network Enclosure Cabinet	Protects, cools, and standardizes hardware mounting.	Mounted switches and patch panels in server room racks.
Patch Panel	Centralizes building cable runs into organized ports.	Terminated incoming structural cabling for easy patch-cord distribution.
Managed Switch	Forwards frames at Layer 2 based on MAC addresses.	Configured portsecurity, access modes, and departmental VLANs.
Punch Down Tool	Seats and trims individual copper wires into IDC slots.	Terminated Cat6 pairs onto patch panel blocks and keystone jacks.
RJ45 Crimping Tool	Crimps modular 8P8C plugs onto twisted-pair cables.	Made custom-length Ethernet patch cables for network connections.
LED / LCD Cable Testers	Tests circuit continuity, pinout mapping, and fault distances.	Verifiedpin continuityand diagnosedcable shorts or split pairs.

CHAPTER FOUR

DURING INTERNSHIP CHALLENGS AND SOLUTIONS

4.1 Major Challenges and Problems Faced

During my internship in the ICT Directorate, I encountered several operational and technical obstacles across database systems, network infrastructure, and system security. First, the existing MySQL database contained years of unvalidated legacy data—such as user names with embedded numbers and special symbols—which made it difficult to introduce strict PHP input validation without breaking legacy report generation. Second, the physical network suffered from poor organization; cables lacked proper labeling, and several hidden UTP runs inside office walls had internal short circuits or split pairs, leading to frequent and unpredictable network drops. Finally, many workstation computers were running outdated, unpatched operating systems with no centralized protection, leaving them highly vulnerable to USB malware infections and recurring system crashes.

4.2 Measures Taken to Overcome Challenges

- **Database Cleansing & Standardization:** Developed and executed a one-time SQL script utilizing UPDATE and REPLACE commands to remove special characters from legacy records, ensuring data integrity without disrupting existing reporting tools.
- **Network Infrastructure Audit & Cable Repair:** Audited the building's network using an LCD wiremap tester, re-crimped damaged RJ45 ends using the T568B standard, and established a clear labeling system for all cables across patch panels and wall outlets.
- **Workstation Security Overhaul & Endpoint Protection:** Performed full data backups, wiped infected hard drives to install clean, fully updated operating systems, and configured Windows Group Policy to disable USB autorun, stopping future malware propagation.

4.3 Results and Discussion

- **Zero Data Corruption (Software & Database):** Implementing PHP RegEx input validation stopped invalid data entry at the source. Database input errors were reported to drop to 0% (include this figure only if supported by test records), allowing the audit team to conduct seamless PIN/TIN searches and error-free fiscal report generation.
- **Network Stability & Speed (Infrastructure):** Resolving cable shorts, re-crimping standard connectors, and configuring VLAN 30 was reported to raise network uptime to

99% (include this figure only if supported by monitoring records). Intermittent disconnects were eliminated, drastically reducing latency when staff accessed the IBEX database.

- **Improved System Performance & Security (Hardware & Endpoint):** Hardware maintenance—such as replacing faulty RAM sticks and power supply units—eliminated blue-screen crashes and improved computer boot times. Clean OS reinstalls and Group Policy restrictions reduced office malware incidents.

4.4 Recommendations Made Regarding Identified Problems

- **Standardize Input Validation System-Wide:** Enforce strict input validation policies (e.g., RegEx rules) across all software modules in every regional branch—not just within the Payroll application—to maintain uniform data integrity and prevent corrupt entries across the entire organization.
- **Implement Automated Off-Site Backups:** Upgrade from manual, localized database backups to an automated, daily off-site backup protocol for the IBEX system. This guarantees robust disaster recovery and protects vital financial data against physical drive failures or localized incidents.

CHAPTER FIVE

OVERALL BENEFITS GAINED FROM THE INTERNSHIP

5.1 Improvement of Practical Skills

- **Software Development & Database Management:** Gained hands-on experience in writing secure, production-ready PHP code to interface with live MySQL databases and perform data validation.
- **Network Engineering & Administration:** Mastered physical cabling techniques—including Cat6 Ethernet termination—and gained practical proficiency in configuring enterprise Cisco network switches using CLI commands.
- **Hardware Diagnostics & Maintenance:** Developed strong hardware skills in building, disassembling, diagnosing, and repairing desktop computer components to ensure operational reliability.

5.2 Upgrading Theoretical Knowledge

- **Networking Concepts (OSI Model):** Applied theoretical concepts from the Physical (Layer 1) and Data Link (Layer 2) layers by terminating UTP cables and configuring MAC-based switch settings on actual network devices.
- **Database Theory (Normalization & Integrity):** Translated database design theory into practice by applying constraints and data cleansing techniques to secure and clean live records in the Payroll database.

- **Theory-to-Practice Integration:** Gained a deeper, intuitive grasp of academic computer science principles by directly observing their real-world impact on system performance and security.

5.3 Improving Industrial Problem-Solving Capability

Through my internship, I developed a structured and methodical approach to diagnosing technical issues within an enterprise IT environment. Rather than relying on trial-and-error, I learned to apply systematic troubleshooting frameworks to resolve complex network and software problems. For instance, when diagnosing a server connection failure, I systematically analyzed the issue from the ground up using the OSI model—first verifying Physical connectivity (checking for cut cables using wiremap testers at Layer 1), then

inspecting Data Link configurations (confirming active switch ports and VLAN assignments at Layer 2), and finally verifying Network settings . Mastering this process of logical deduction significantly enhanced my technical problem-solving capabilities and taught me how to resolve critical system outages efficiently.

5.4 Improving Teamwork and Interpersonal Skills

Working within the ICT Directorate reinforced the importance of clear communication and active collaboration in a fast-paced technical environment. Tasks such as mounting network cabinets, running structured cabling, and managing server infrastructure required seamless coordination with senior IT staff. I learned how to effectively divide technical workloads—such as pairing up so one team member tests wall outlets while the other patches connections at the server rack—to complete installations safely and efficiently. Additionally, this experience helped me develop strong active listening skills, accept constructive feedback from experienced colleagues, and align my daily tasks with the broader operational goals of the IT team.

5.5 Improving Leadership and Project Management Skills

As my technical competence and familiarity with the office infrastructure grew, I was granted the autonomy to lead the operating system upgrade project across the finance department's workstations. Taking ownership of this initiative required me to develop schedule management skills, carefully coordinating PC downtime during off-peak hours to minimize operational disruption for the finance staff. Successfully directing this deployment—from initial planning and data backup to software installation and user handoff—significantly enhanced my leadership, time management, and project execution capabilities.

5.6 Understanding Work Ethics and Industrial Psychology

My immersion in the ICT Directorate deepened my appreciation for professional work ethics, specifically regarding data privacy and confidentiality. Working within a finance office granted me elevated administrative access to highly sensitive records, including payroll details and taxpayer information. This responsibility reinforced the critical ethical standard of maintaining absolute data confidentiality and strictly refraining from accessing or modifying records beyond my authorized duties. Furthermore, I observed industrial psychology principles firsthand by recognizing the high-stress nature of the finance workplace during technical outages. Experiencing how system downtime impacts non-technical staff taught me

to respond to IT help desk requests with empathy, clear communication, and patience, ensuring that technical support was delivered professionally while easing user frustration.

5.7 Entrepreneurship Skills Gained

My experience at the ICT Directorate broadened my perspective regarding the commercial and entrepreneurial potential within the IT industry. Through direct hands-on experience, I recognized that skills such as structured network cabling, hardware diagnostic and repair, system optimization, and database administration are in high demand across local businesses, academic institutions, and healthcare clinics. Gaining practical mastery in these areas demonstrated that delivering reliable

technical services requires relatively low capital investment—primarily essential diagnostic tools, specialized hardware equipment, and technical expertise. Consequently, this realization has provided me with the practical foundation, self-reliance, and business acumen necessary to launch an independent IT consultancy and network services enterprise following graduation.

5.8 Improving Interpersonal Communication Skills

As an IT intern, communicating technical concepts to non-technical staff proved to be a daily operational necessity. I learned the importance of simplifying complex technical jargon into clear, accessible language when assisting office personnel. For example, when finance managers inquired about network latency, rather than explaining abstract networking terms like broadcast domains or packet collisions, I communicated the issue by explaining that the network was experiencing heavy traffic and that configuring a dedicated virtual network (VLAN) created an exclusive, high-speed path for critical payroll data. Mastering the ability to translate technical concepts into plain language bridged the communication gap between the ICT Directorate and other departments, fostering greater trust, mutual respect, and smoother collaboration across the entire organization.

CHAPTER SIX

GENERAL CONCLUSION AND RECOMMENDATIONS

6.1 General Conclusion

The internship program at the Sekela Woreda Finance Office served as a transformative milestone in my academic and professional development, successfully bridging the gap between theoretical Computer Science concepts learned at Debre Tabor University and the practical demands of an enterprise IT environment. Through active involvement in optimizing the office's IT infrastructure, I gained the technical competence and confidence required to transition into a full-fledged IT professional. On the software engineering front, resolving critical data-entry vulnerabilities in the Payroll system using PHP Regular Expressions permanently eliminated database corruption and ensured data integrity. On the hardware and networking front, hands-on tasks—including Cat6 cabling, patch panel termination, Cisco switch VLAN configuration, and hardware diagnostics—directly enhanced the stability, security, and performance of daily financial operations.

Beyond technical achievements, this experience cultivated essential professional soft skills necessary for a successful career in technology. Working alongside senior staff enhanced my ability to collaborate within a technical team, translate complex concepts for non-technical users, troubleshoot under pressure during network outages, and maintain strict data confidentiality when handling sensitive financial records. Ultimately, receiving consideration for future employment at the conclusion of the internship served as a meaningful validation of the practical impact and value delivered by the technical solutions implemented during this program.

6.2 Recommendations for the Hosting Organization

Based on the technical assessments and operational challenges observed throughout the internship, several strategic measures are recommended to ensure the long-term security, stability, and efficiency of the Sekela Woreda Finance Office's IT infrastructure. First, the organization should transition from manual file-saving protocols to an automated, encrypted off-site backup system for all IBEX and Payroll MySQL databases, safeguarding critical financial records against catastrophic data loss caused by hardware failure, localized disasters,

or physical damage. Second, to enhance endpoint security and reduce administrative overhead, the ICT Directorate should deploy a centralized patch management framework, such as Windows Server Update Services (WSUS), to automate the distribution of operating system patches and antivirus definitions across all workstation computers simultaneously.

Additionally, to maintain network organization and simplify future troubleshooting, the ICT Directorate should enforce a strict cable management policy requiring that any new network connection be systematically labeled at both the client outlet and patch panel ends, preserving the structured network baseline established during this project. Finally, the organization should conduct quarterly IT security awareness training for finance and audit staff. Educating end-users on safe internet practices, social engineering threats, and proper data-entry procedures will minimize malware risks and further protect the integrity of the organization's digital assets.

6.3 Recommendations for Future Interns

To maximize the educational and professional value of the internship experience, several key recommendations are offered for future Computer Science students. First, students should solidify their core technical fundamentals prior to starting their placement. Having a strong practical grasp of SQL query writing, core programming logic—such as PHP Regular Expressions—and basic networking principles like IP subnetting and Cisco CLI configuration enables interns to contribute meaningfully to live IT tasks from day one. Second, interns should demonstrate high levels of initiative and proactivity rather than waiting passively for assigned duties. Identifying operational bottlenecks, such as an unorganized patch panel or an under performing workstation, and proposing solutions to supervisors demonstrates drive, self-reliance, and problem-solving capability.

Finally, future interns must cultivate effective communication and recognize the importance of asking for guidance when uncertain. Working on live enterprise infrastructure, central databases, and mission-critical production servers carries significant risk; executing an unverified script or altering a core switch configuration without confirmation can lead to system-wide outages. Seeking clarification from experienced IT staff ensures operational safety, prevents costly errors, and accelerates the intern's practical learning curve throughout the program.

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APPENDICES

This section breaks down the main hands-on tasks and solutions completed during the internship at the Sekela Woreda Finance Office into simple, easy-to-understand explanations.

Appendix A: Fixing the Data Entry Problem (Software Validation)

The Problem: Before I started, the office payroll and IBEX software allowed anyone to type numbers or symbols (like `@`, `#`, or `$`) directly into name fields. When these invalid symbols ended up in the system, you corrupted the database and caused unexpected crashes during financial audits.

What I Did: I added background validation checks (using pattern-matching rules called Regular Expressions) to automatically inspect what a user types into a form before saving it.

- **How It Works:**

- **Automatic Check:** When a staff member types an employee's name, the system instantly verifies that it contains only actual letters, spaces, hyphens, or apostrophes.
- **Instant Error Warning:** If an incorrect symbol or number is entered, the system blocks the submission and displays a clear message asking the user to fix the entry.
- **Data Protection:** This keeps the database clean, prevents software crashes, and protects payroll records from permanent errors.

Appendix B: Employee Quick-Search Tool (PIN & TIN Lookup)

The Problem: Finance officers previously spent hours sifting through physical paper files or running slow database checks just to confirm an employee's tax details during monthly processing and annual audits.

What I Did: We built a streamlined, instant search feature directly connected to the office's MySQL database.

- **How It Works:**

Simple Input: A staff member enters an employee's unique **PIN** (Personal Identification Number) or **TIN** (Tax Identification Number) into a single search bar.

Instant Lookup: The system immediately retrieves and displays the employee's full profile—including their department, current salary rate, and tax status—in seconds.

Time Saved: This removed manual paperwork, cut auditing time significantly, and eliminated lookup mistakes.

Appendix C: Network Reorganization & Security Setup

The Problem: The network room had loose cables hanging everywhere, computers constantly lost connection, and all departments shared a single open network—exposing sensitive financial data to unnecessary security risks.

What I Did: We rebuilt the office's physical cabling setup and separated the network traffic using managed switches.

Step-by-Step Summary:

- **Step 1:Cabinet Installation:** I wall-mounted a central network rack cabinet to lift key equipment off the floor, keeping switches clean, cool, and secure.
- **Step 2:Organized Cabling (T568B Standard):** I trimmed Cat6 network cables to exact lengths and punched them down into central patch panels. This organized all incoming building lines so cables wouldn't tangle or disconnect when pulled.
- **Step 3: Department Isolation (VLAN 30):** I logged into the main Cisco switch and created a dedicated virtual network (**VLAN 30**) specifically for the Payroll Department. This isolated payroll communications from general office traffic, improving data privacy and eliminating network slow-downs.

Appendix D: Weekly Internship Activity Log

Below is a week-by-week timeline summarizing the key milestones completed over the two-month training period:

Table D.1 Weekly Internship Activity

Period	Major Tasks Completed	Assigned Unit
Weeks 1–2	Surveyed office hardware and cabling, interviewed finance staff, and diagnosed why the IBEX software was crashing.	ICT Directorate & Finance Office
Weeks 3–4	Programmed input validation rules in PHP to block special characters and built the SQL quick-search tool for PIN/TIN lookups.	Software Development Unit
Weeks 5–6	Serviced 35 workstations: cleared physical dust, replaced failing power supplies and RAM, wiped malware, and installed clean Windows OS builds.	Hardware Maintenance Lab
Weeks 7–8	Mounted the server room network rack, terminated Cat6 cables into patch panels using the T568B standard, configured Cisco VLANs, and handed over the system.	Network Infrastructure Unit